

Chapter 6

Regional Network Analysis

6.1 Introduction to Network Modelling

Explaining dynamic past behaviour using data from a *static* archaeological record is a challenging inferential task. The use of computer modelling and quantitative methodologies can help reduce the uncertainties resulting from these challenges. For example, Chapter 5 examined in detail 36 potential landuse scenarios at the site of Tell esh Shuna North for each of the three periods being examined in this thesis (Chalcolithic, EEBI and LEBI). The primary focus of this research has been presented in Chapter 5, and although the results of that analysis proved useful in many ways, the fact that Shuna North was examined in isolation limited the insights into extent and location of landuse, degree of impact and visibility of land modification in EB I in the area. In a number of places in the preceding chapter, it was suggested that further exploration of the results to include neighbouring sites could affect the assessment of a scenario or the end conclusions resulting from the modelling process in general. Chapter 6 therefore represents a natural continuation of an isolated settlement in Chapter 5, building on the findings by considering the implications of incorporating subsistence catchments generated for other communities in the region.

6.1.1 Aims

As discussed in the preceding chapter, neighbouring settlements have the potential of limiting access to crop and grazing land. Territorial expansion, or in other words the utilisation of more land to increase food production, could be limited by many factors, including an unwillingness to travel distances beyond a particular threshold, environmental constraints, or as stated above, lack of available land due to use by other communities. Regardless of the cause, if territorial expansion was not possible, increases in population levels could not be possible without changing subsistence strategies. As shown in Chapter 5, it is possible to lower land requirements whilst increasing food production.

This chapter looks at Shuna in conjunction with 125 other EB I sites to illustrate how insights into some of the following questions can be addressed:

- Regionally, how much land may have been under cultivation?
- How much land in the region had to be dedicated to use as grazing land?
- Did land requirements for the settlements in the region indicate possibilities for conflict over competition for land?
- If the possibility for conflict is indicated, was this more prevalent in grazing land needs or cultivated land needs?
- Do larger settlements create particular pressures on their local landscapes?
- Is the clustering of large sites problematic?
- Do certain scenarios create more difficult circumstances which make it possible to question some combinations of diet/management practices?
- Were there areas where farmers and shepherds probably had to negotiate more intensively for access to land, and where this access might have strained relationships with neighbouring settlements?
- Were there areas where land demarcation through field boundaries, terraces etc. might have been more important?
- How do routes relate to intensively exploited areas? Do they pass through them, making clear to passers by the question of control and ownership?
- What is the effect of natural barriers such as lakes and rivers on conflict potential resulting from competition for access to land?

The remainder of this chapter presents a strategy which shows how Landuse Analyst can be incorporated into a regional study to address questions like those above. Computational requirements for this type of analysis are enormous when done on a regional level, and given the time constraints imposed on this research, it has limited the level of analysis possible to be presented in this chapter. The analysis that has been possible, however, clearly shows the implications and potential of this work.

This chapter also compares different territory allocation methods (e.g. Thiessen polygons, XTent models, Landuse Analyst catchments) in a limited region to the south and west of Shuna. This region was selected to illustrate territory allocation for two reasons:

- The Digital Archaeology Atlas of the Holy Land^{6.1} (DAAHL) provides an online database of several hundred archaeological sites throughout the southern Levant, including 627 EBA I sites. Sites in and east of the Jordan valley in this database give approximate site sizes which provides an avenue for estimating population levels.
- the ASTER tile, used to provide elevation data, conveniently encompasses 126 EBA I sites (including Shuna) that have size estimates which provides the basis for making estimates of population sizes (a requirement when using Landuse Analyst to calculate subsistence catchment areas).

In addition to this comparison of territory allocation methods, preliminary work investigating the potential nature of site connections/interaction based on potential walking routes is presented. The route networks use all 627 EBA I sites in the DAAHL database and calculates the routes using elevation data from Shuttle Radar Terrain Mapping (SRTM) between each site, optimised to ‘prefer’ settlements en-route as way-points.

In an effort to better understand the impact and implications of landscapes surrounding Shuna, the 125 additional sites that are dated to the EB I in the DAAHL database are now analysed in relation to their landscape footprints. Time constraints have meant that it has not been possible to examine these sites in the same detail as Shuna, therefore estimates on population and contemporaneity have been based on data taken directly from the DAAHL database. The results of this analysis are to be taken with a certain degree of caution; the ‘averaged’ diets reconstructed for the populations was used as a standard for all sites, and the contemporaneity/size of the sites as recorded in the database were not independently investigated. Even with the coarse nature of the input data for this analysis, the results clearly indicate areas of interest and insight and prove useful for extending the analysis done in Chapter 5.

It is pertinent to note that, given the coarse nature of the input data (the elevation data used is either ASTER data or SRTM data, 30 m and 90 m resolution respectively), modern terrain features such as roads, canals and so on will have virtually no effect (if any) on the analysis done. In addition, as will be explained later in this chapter, the analysis is optimised to use known EBA sites as waypoints when calculating routes, with pre-Modern Ottoman travel routes providing controls on any influence that recent terrain modifications might have.

6.2 Input Variables for Network Analysis

An EBA I standardised diet has been assigned using Shuna’s EEBI diet. The decision to use the EEBI diet rather than the LEBI diet was based on the drastic change in the

^{6.1} available online at <http://daahl.ucsd.edu/DAAHL>

faunal assemblage at Shuna evidenced in the LEBI. In part, this decision was also based on the fact that the size estimate for Shuna in the DAAHL database matches the EEBI estimates used in this thesis. The question of pigs contributing to the diet at other sites might be problematic, as Shuna does have a high number of pigs in the assemblage, but after adjusting the input values in the analysis given by Grigson (1995), it can be seen that in Shuna's region (Area 'C' in her study) the average levels of pig in the Chalcolithic is over 20% (contributing more than sheep and goats combined) and over 12% even in the EBA. When, using her data, the EB IV settlements are graphed independently, it shows that region-wide, pigs remain more important than caprines, and still contribute nearly 20% of the meat (Figure 6.1. This trend is also present at Shuna (Figure 6.2).

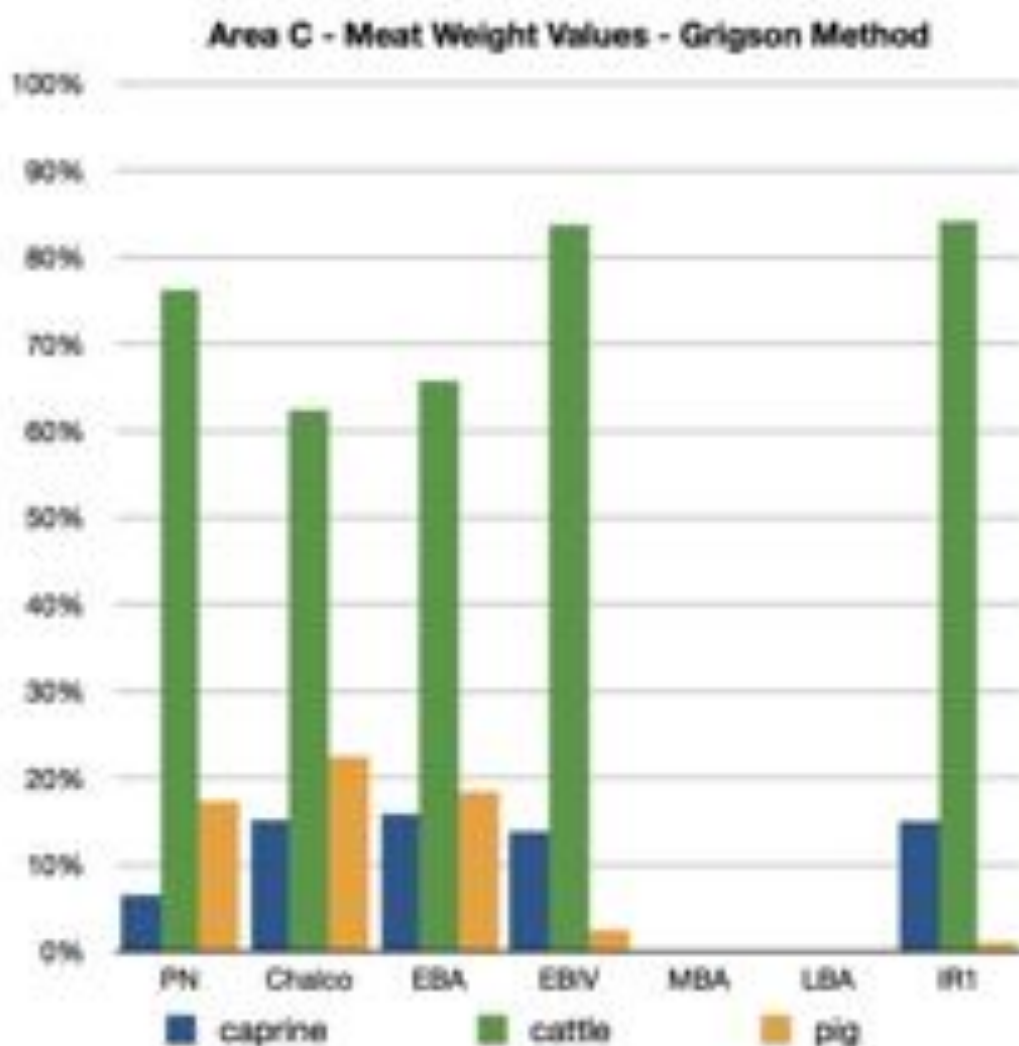


Figure 6.1: Animal contributions: Grigson Area C. After Grigson (1995) with settlements from EB IV (her data) shown separately.

Having looked at the results of the modelling done in Chapter 5, it appears that the

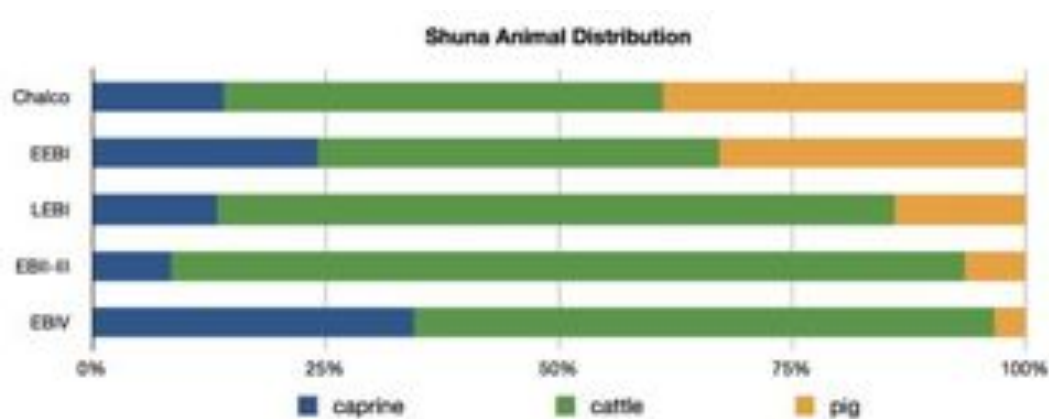


Figure 6.2: Animal contributions at Shuna.

steadily declining importance of pigs throughout the EBA could be a result of a change in subsistence strategies, specifically one which increased reliance upon cereals and dairy whilst lowering the reliance on meat; the same association has been suggested in other periods by others (e.g. Redding 1991, Hesse and Wapnish 2001).

As discussed in Chapter 5, as populations grew, the calculated land requirements for the EEBI diet suggest that this change would have been beneficial, and possibly necessary, to accommodate an expanding population. In order to investigate factors that might have contributed to a change in animal management between EEBI and LEBI at Shuna, scenarios were modelled using the EEBI diet, in particular it was hypothesised that a high meat element in the diet may have been a contributing factor. Population levels have been calculated using the same method used in Chapter 5; area of the site multiplied by estimated population density. Site size was provided by the DAAHL database and population densities of 100 and 300 individuals hectare⁻¹ are used to determine high and low population estimates.

6.3 Thiessen Polygons and XTent Modelling

Thiessen polygons are a convenient way of tessellating a landscape; beginning with points, which in this case are EB I archaeological sites, connecting lines are made between each point and its nearest neighbours. These connecting lines are then bisected to form boundary lines. The boundary lines are extended to form polygons which represent ‘territories’ for each point. As Savage and Falconer (2003, 32) point out however, Thiessen polygons “...invariably centres polities around a major settlement, reifies an assumption of firm boundaries, and overlooks surrounding networks of smaller communities.”. In other words, they fail to take into consideration the idea that smaller neighbouring settlements may have been part of the same polity as the large site. Thiessen polygons give equal consideration to every point, whether they are

20 hectare settlements or a one hut summer camp.

Renfrew and Level (1979) developed the Xtent model, which considers the influence of a settlement when calculating territorial boundaries, to overcome the above outlined limitations of Thiessen polygons. The Xtent model assumes that the influence of a site is a function of the relative size of the site and its distance from other sites ('points of interest') or stated more succinctly, that "... the size of the territory controlled by a centre is proportional to the site's importance" (Hare 2004, 802). Xtent models incorporate the smaller sites which are located in the vicinity of larger ones, and are assumed to be under the control of these larger settlements.

Thiessen polygons were generated for the 126 sites in the selected study area based on the ASTER tile extents. Using the ASTER tile for elevation data, the results were overlaid on the results of XTent modelling. The XTent modelling was done in GRASS (using Ben Ducke's yet to be released *rxtent* module^{6.2}, which he kindly provided to me) which is a full implementation of Renfrew and Level's Xtent model. The XTent modelling uses site size from the DAAHL database as the weighting factor (Thiessen polygons do not use any weighting factors), and results in some of the smaller sites being wrapped into the territories of the larger ones. The results of this analysis is shown in Figures 6.3 and 6.4. As evident in these two methods, there is an increasing degree of sophistication from Thiessen polygons to the XTent modelling results. It is possible to use the catchments generated with Landuse Analyst to add further sophistication to the XTent model. By generating all 126 catchments simultaneously and not allowing overlap, a very robust model would emerge that would show territories based on subsistence needs and time of travel across the landscape. The development of the required algorithms in conjunction with the extremely high computational load meant that it was not possible to fully implement this methodology given the time constraints afforded this research. Instead, catchments are generated individually, and the results compiled to show where areas of overlapping landuse occur. These results can be viewed in conjunction with the Xtent results to look for alternate land catchment placement for sites with overlapping catchments (see Figures 6.12 6.19). In order to give a better indication of where people might have been moving through the catchments, the following section explains how the regional path networks are created. The section on path networks provides a better understanding for the catchments discussed above, and the section following the network paths explains how the catchments were created, and presents the results in conjunction with the path networks.

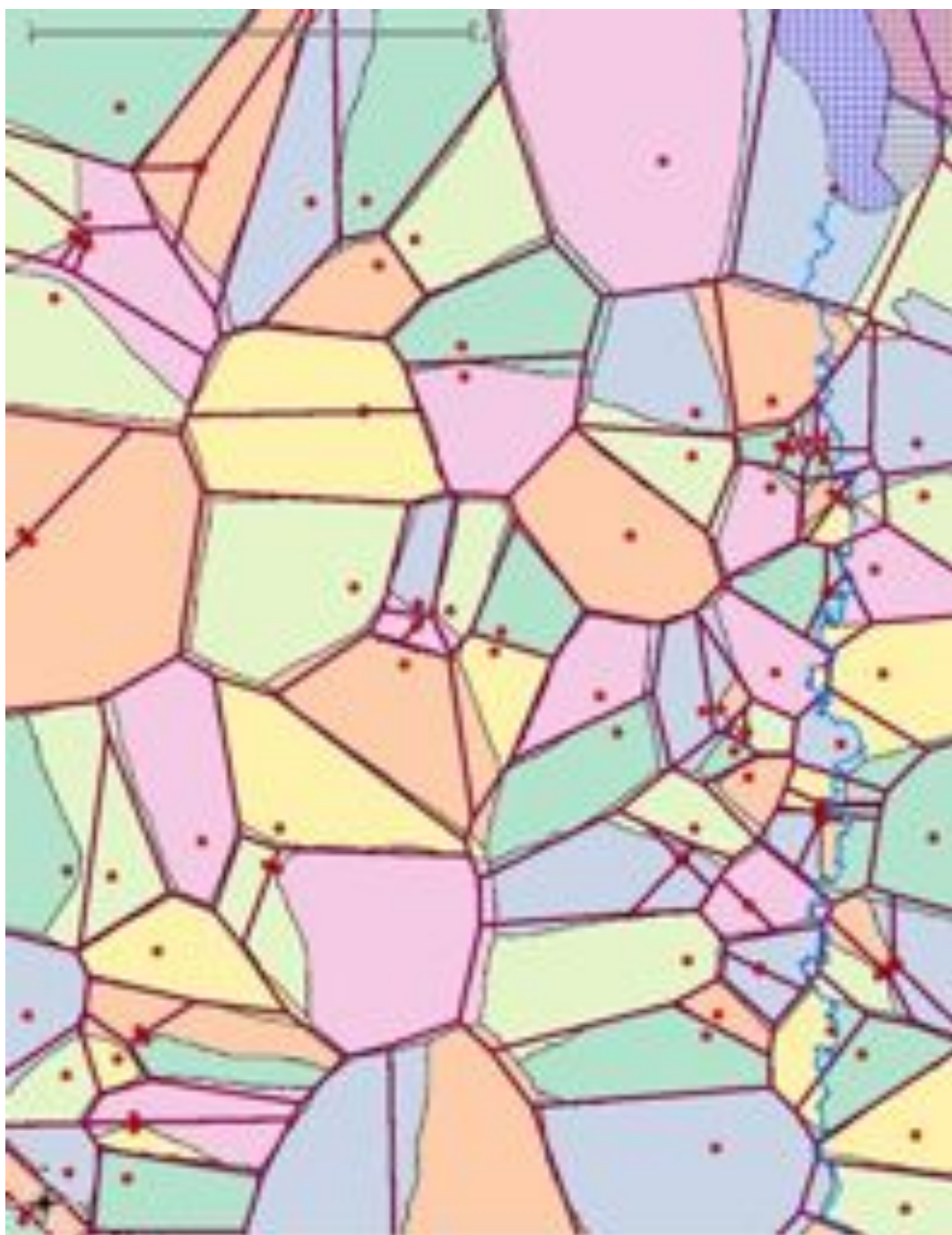


Figure 6.3: XTent modelling versus Thiessen polygons. XTent shown as coloured polygons, Thiessen polygons are outlined with thick magenta lines.

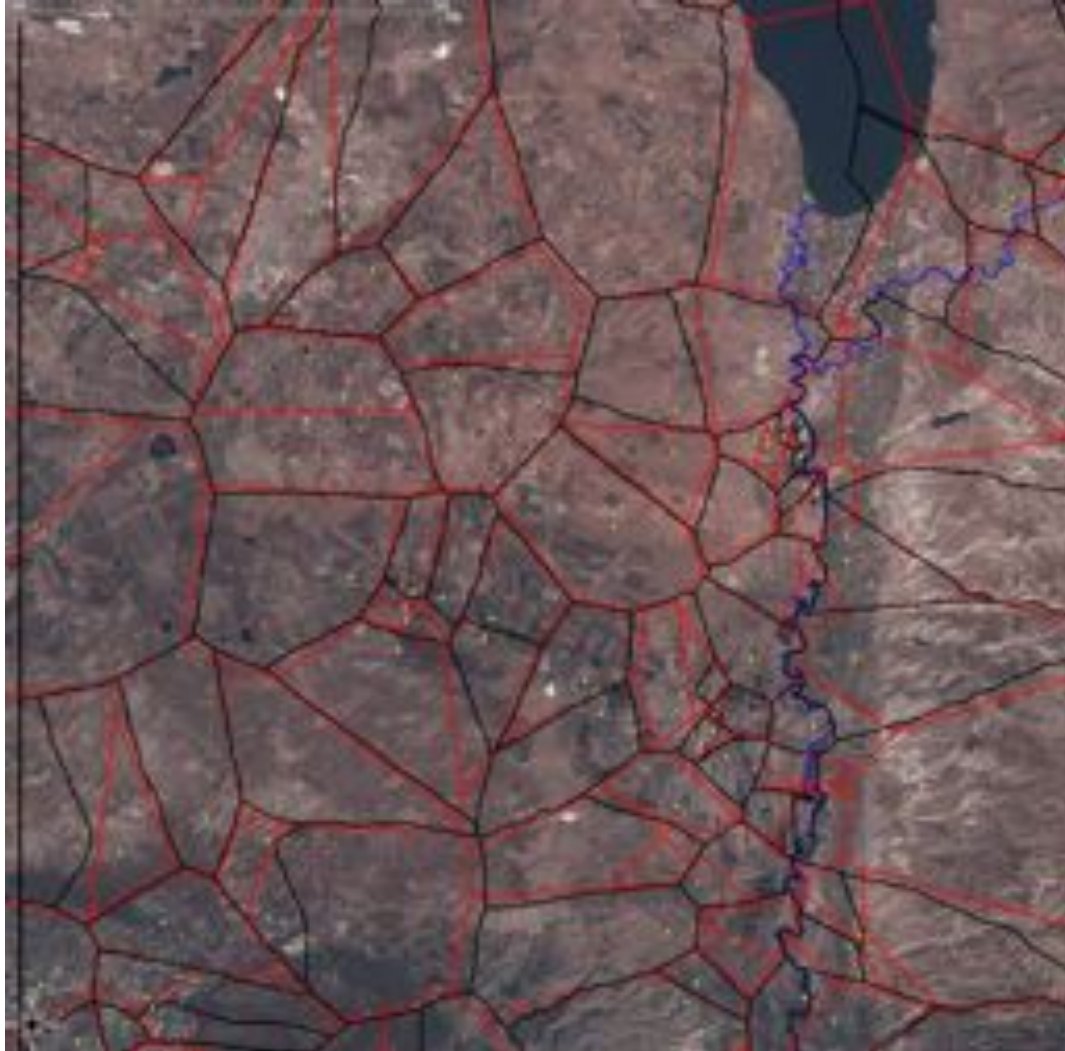


Figure 6.4: XTent modelling versus Thiessen polygons. XTent shown with black lines, Thiessen polygons shown with red lines.

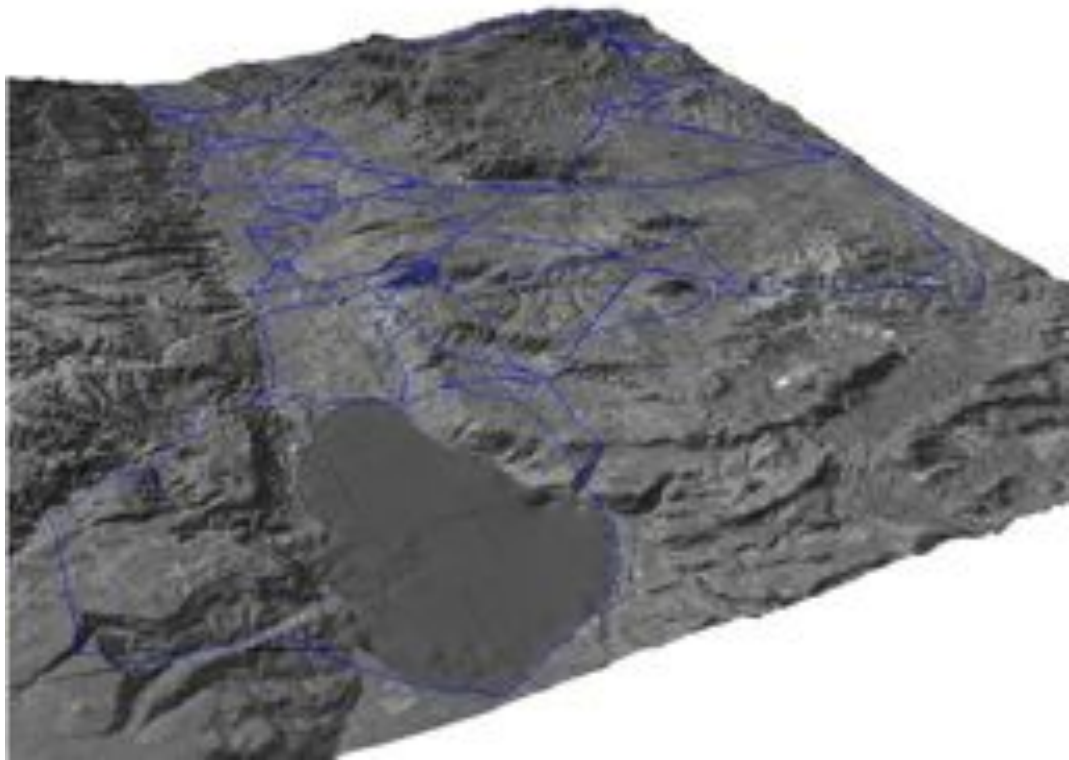


Figure 6.5: Looking SSW at the path network

6.4 Path Networks

Calculating the potential network of paths connecting sites requires several steps and finding walking routes which favour other settlements as ‘way-points’ takes a great deal of computation; the results presented here took several days to compute running continuously and simultaneously on five Intel Quad-Core computers, each with 4GB of ram. The hope is, however, that this work will be possible to do in the future at the same resolution as the analysis which was done in Chapter 5, and provide a means to examine trade potential between sites based on production potential of different commodities at each site.

The regional site catchment strategy presented above is, as was discussed, not done to the same degree of detail as Shuna; likewise, when the regional level is expanded to include the entire southern Levant, the level of detail again must be lowered. Elevation data (which is the basis for generating the cost surfaces upon which walking routes are calculated) used in the analysis of 126 sites in conjunction with site catchment modelling was provided by an ASTER tile, which has a resolution of 30 metres; SRTM is 90 metres. To analyse the entire southern Levant at this resolution is possible, but

^{6.2}xtent is an Open Source module written for use with GRASS which was compiled against GRASS version 6.3 source code.

the computing time to do it would be enormous. This issue is discussed further in the final chapter of this thesis.

When using an anisotropic cost surface based on walking time, the best route from A to B can be different than the best route from B to A (think of yourself cross-country skiing, going from the top of a valley to the bottom of the valley “A to B”; now consider the opposite journey, from the bottom to the top “B to A”). Because of the anisotropic nature of the routes, walking paths need to be generated in a particular fashion; each site must be used as the starting point for routes which end at every other site included in the analysis. In this case (using 126 sites total) each site in turn acts as the starting point for journeys to the other 125 sites. Running this analysis without using way-point optimisation results, therefore, in $125 \cdot 126 = 15,750$ routes; Figure 6.7a shows only partial results of running the analysis this way. Clearly, this method is unsuitable. Figure 6.7b shows the results using way-point optimisation and clearly illustrates the benefits of the way-point optimisation method.

Step one in this process creates a cost surface based on walking time for the entire region. This is done using the *r.walk* GRASS module. By designating every site as a starting point rather than just one, a multipoint cost surface is created (see Figure 6.6).

The next step in the process generates individual cost surfaces (again with *r.walk*) using the co-ordinates of each site as a starting point. Crucial in this step is including the multi-point cost surface (generated in the first step) as a friction surface. By doing this, additional cost is assigned to areas further from a site. Thus, the closer you travel to other sites, the less friction is added while en-route. This is how preference is given to travelling through sites, treating them as ‘way-points’. In this case, 126 sites were used, and 126 individual cost surfaces were generated. Once each site has its own walking time cost surface created with the multi-point cost surface used as friction, the paths are calculated using the *r.drain* module. This module allows the user to enter the starting co-ordinates, and supply a list of target co-ordinates. Therefore, the command is run once for each point of origin (or in this case 126 times) and given the co-ordinates of every other site (in this case 125) and results in a raster map showing the calculated routes from the origin to each of the other sites. These result rasters are accumulated into one final results raster (Figure 6.7b) so that each time a path is re-used, its value increases. This allows us to see which paths are being used by the most sites. The output raster can be converted into a vector, preserving the accumulated values, and grouped into different categories. For this example, the values of Lightly Used, Well Used, and Heavily Used were chosen. Useful as well when viewing these paths is to look at the site sizes (see Figure 6.11).

As seen in Figure 6.8 on page 333, the more rugged land (e.g. the bottom left quadrant of the image) has no travel routes or settlements. The result is that the territories created with the Xtent modelling (Figure 6.8a in yellow) are much larger. Because of

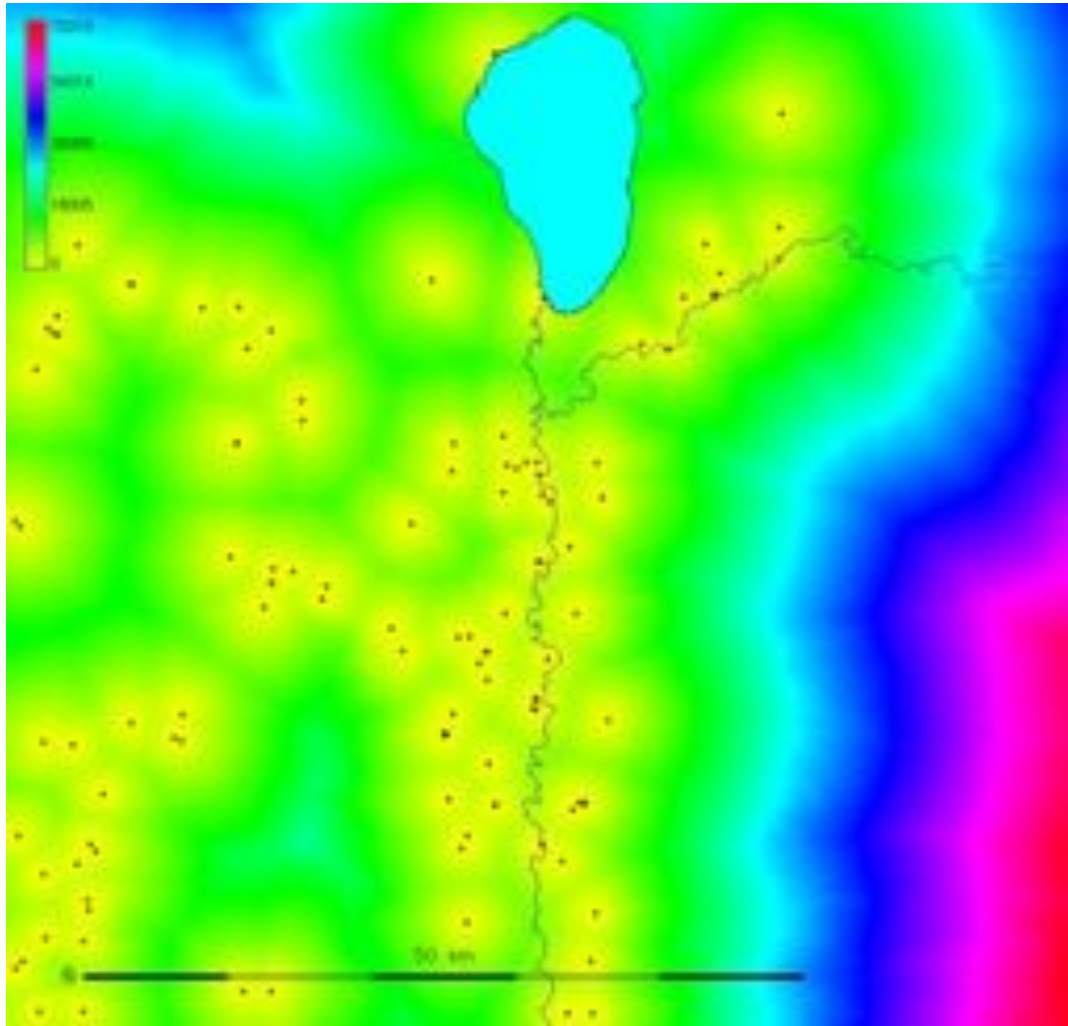


Figure 6.6: Cost Surface based on walking time with multiple origins. Legend shows relative travel cost value.

the rugged nature of the landscape, this land would have been more suitable for grazing animals than cultivation (this is evident from the satellite imagery used as background in the image, with little or no cultivated land visible in this area) giving the settlements that are associated with these Xtent territories access to a great deal of grazing land. This site deficiency could be a function of a dearth of survey, but in the absence of better data, it seems reasonable to assume that at a minimum, these areas were less densely populated. It is important to discuss this when looking at routes, because as seen in the images, there are few routes that go through these rugged regions. The lack of routes would provide a certain amount of security via isolation, which might indicate that there is a correlation between security needs and subsistence strategies, but ‘meatier’ subsistence strategies have also been shown to potentially limit population levels. Regardless, at a minimum, differences in subsistence strategies could be expected in areas like this, especially if there is suspected regional land pressure limiting

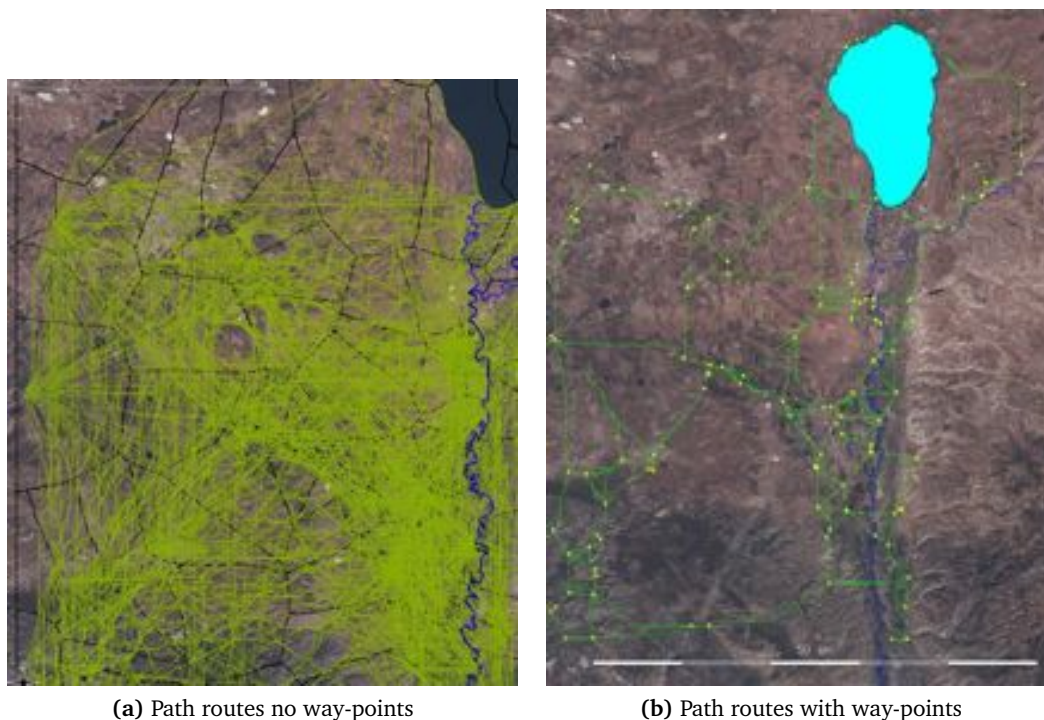


Figure 6.7: Examples of route calculations with and without optimisation for using en-route settlements as way-points. Clearly illustrated are the single line routes between sites (on the right) versus the mass of tangled lines on the left. *Route calculations are shown in detail in subsequent images, and shown in this small form here for illustrative purposes only.*

the expansion of cultivated land. Furthermore, these areas would provide more security to shepherds due to the lower likelihood of chance encounters with unfriendly people who may have wanted to steal their animals for their own purposes (as discussed in Section 5.8.1 on page 317). Further advantages to the way-point optimised walking time path route method are clear when presented with Delaunay analysis, which does not take topography, time or way-points into consideration when calculating the route network. Delaunay tessellation (closely related to Thiessen Polygons) can be used in archaeology to define site territories (Savage 1990, e.g.), and Delaunay triangulation is a way of triangulating sites to form a network which can be used for shortest route type analysis. This type of analysis is useful for things without geographical factors (such as in the design of computer networks), and is clearly less valuable for use in archaeological investigation than methods which account for the natural terrain upon which people travelled. It is interesting to view how some general trends appear in both types of route calculations, however, particularly evident in more densely populated areas like the Jordan Valley. It might, in fact, be useful to look at the triangulation method as being suggestive of relationships between communities rather than routes. In this light, a combination of the two methods can provide clues as to what the true nature of regional site interactions and relationships might have been.

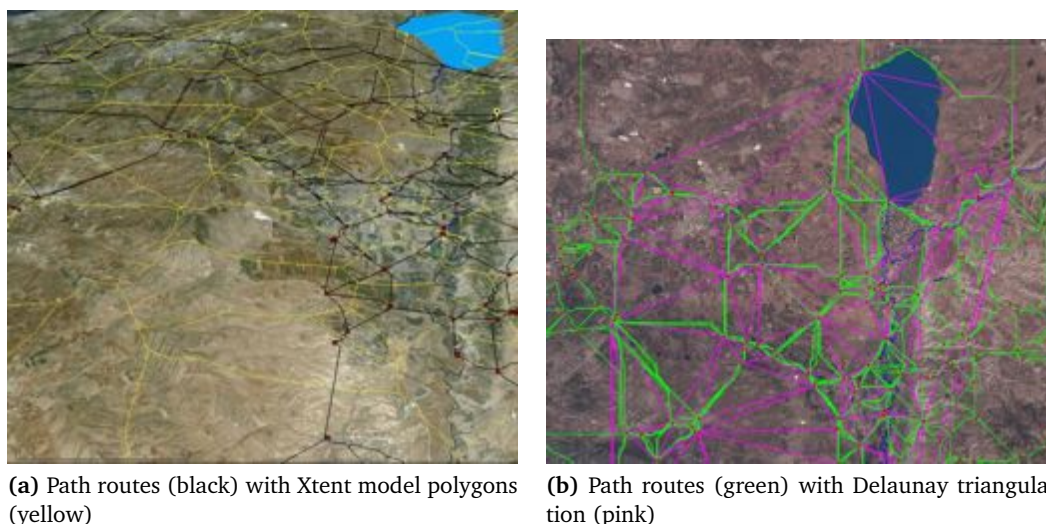


Figure 6.8: Comparing path route networks with Delaunay triangulation and Xtent model polygons.

When viewed on a DEM in 3d, the path network, shown in Figure 6.5, clearly shows a preference for both settlements and less rugged terrain, which is exactly the desired result. Figure 6.9 on page 334 shows the final results of the path network analysis for the entire Early Bronze Age I southern Levant, loaded into Google Earth.

The impression of a community on people travelling through the area would have been highly influenced by at least four things:

1. The size and nature of the settlement itself.
2. The amount of land which was visibly identifiable as having been modified by the settlement (e.g. cleared and/or cultivated).
3. Deliberate modifications intended to convey a statement of their presence in the region (e.g. dolmens, cairns, cemeteries).
4. The presence of (and density of) people.

By looking at where these routes may have been located, insights can be made which will enable these kinds of questions to be made. For example, if dolmen fields can be shown to have been visible from routes that connected certain settlements, it can imply that they were so placed to intentionally make people travelling these routes aware of their presence. This takes us back to one of the questions presented earlier in this thesis, which was whether people were intentionally inscribing themselves on their landscapes.

No doubt, some of these calculated routes may not have been used, and additional routes will also certainly have existed. This method would benefit from enhancements

that could generate direct routes without way-points between larger sites. Insights into this issue can be made from an examination of known, historical trade routes.



Figure 6.9: Path network based on travel time optimised for using en-route settlements as way-points connecting 627 EBA I settlements as listed in the DAAHL database. SRTM used for elevation data. Shuna is indicated with the yellow pushpin.

Esse (1991, 2) produced a map of important trade routes in the southern Levant based on historical Ottoman texts which have been digitised and shown together with the routes generated using the methods described above (Figure 6.10). The results show clear trends as far as identification of general route ‘corridors’ (e.g. through the Beth Shan valley).

The first four of eight maps depicting weighted path networks, EB I sites (sized), results of XTENT analysis, and catchment analysis results for EEBI catchment extents are shown in Figures 6.16 to 6.19.

6.5 Regional Catchment modelling

In addition to Shuna, 125 other sites were chosen for analysis using the extents of the ASTER tile as a boundary within which all EBA I sites from the DAAHL database were selected. The selected sites were then extracted from the database, and used for catchment modelling. The co-ordinates for each site and the associated site sizes were then used for modelling with Landuse Analyst. Each individual site was modelled manually (i.e. the co-ordinates and population level had to manually entered for each



Figure 6.10: Ottoman trade routes (white) compared to way-point optimised travel time routes. (B)eth Shan, (M)egiddo, Khirbet (K)erak and tell (S)himron indicated on map, as well as Shuna with the yellow pushpin.

model run) because the *Experiment* module for Landuse Analyst has not yet been implemented^{6.3}. The resulting catchments were then compiled into one map with GRASS's *r.mapcalculator*. The benefit of sequentially adding each of the generated catchments together with *r.mapcalculator* is that any areas where the catchments overlap increases the value of that area by one. If there are additional overlapping catchments, this value increases sequentially by one, so that when all 126 maps have been combined, it is possible to see how many sites were potentially competing for certain areas (e.g. if an area has a value of 3, three sites have had catchments generated which include this land).

The results of the consolidated catchments (separated into cropland/grazing catchments for both minimum and maximum population levels) with the XTent model results are shown in Figures 6.12 to 6.19. Included in these figures are the results of the path network modelling, discussed in Section 6.4.

^{6.3}the Experiment feature of Landuse Analyst will allow the user to simply select a data file, like the one described above, hit *GO*, and walk away



Figure 6.11: Regional path network showing EB I sites in and west of the Jordan valley. Site sizes are indicated as well as relative usage of the paths. It should be noted that path usage indicates the number of sites utilising them, not the amount of ‘traffic’ on them.



Figure 6.12: Regional map showing weighted path networks, EB I sites (sized), results of XTENT analysis, and catchment analysis results for EEBI minimum population maximum crop extents smallest footprints scenario. Light green indicates land desired for use by one settlement. Dark green indicates land desired for use by 2 settlements. Red and Yellow indicate several sites desiring the same land.



Figure 6.13: Regional map showing weighted path networks, EB I sites (sized), results of XTENT analysis, and catchment analysis results for EEI maximum population maximum crop extents smallest footprints scenario. Light green indicates land desired for use by one settlement. Dark green indicates land desired for use by 2 settlements. Red and Yellow indicate several sites desiring the same land.

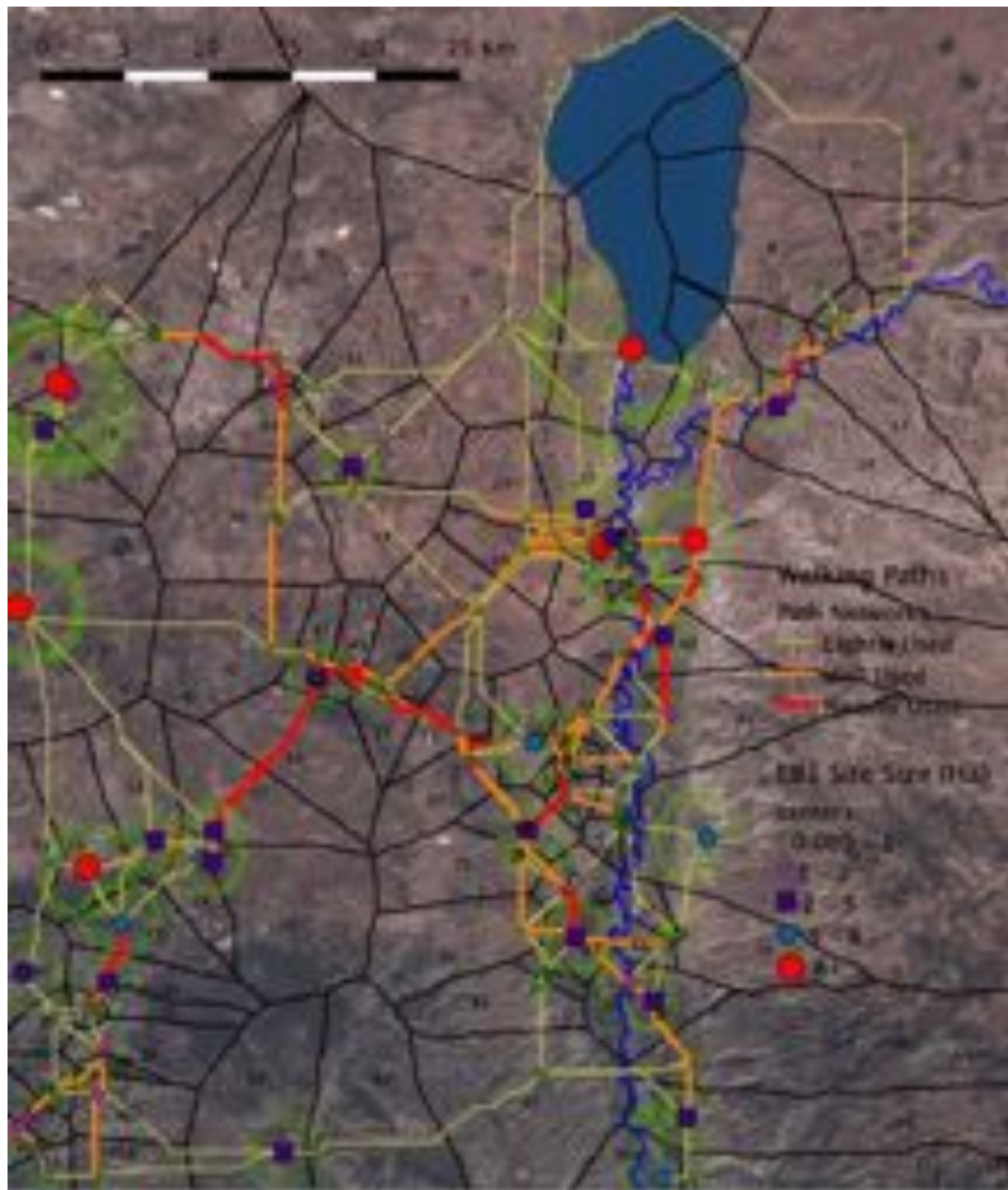


Figure 6.14: Regional map showing weighted path networks, EB I sites (sized), results of XTENT analysis, and catchment analysis results for EEBI minimum population maximum grazing extents smallest footprints scenario. Light green indicates land desired for use by one settlement. Dark green indicates land desired for use by 2 settlements. Red and Yellow indicate several sites desiring the same land.

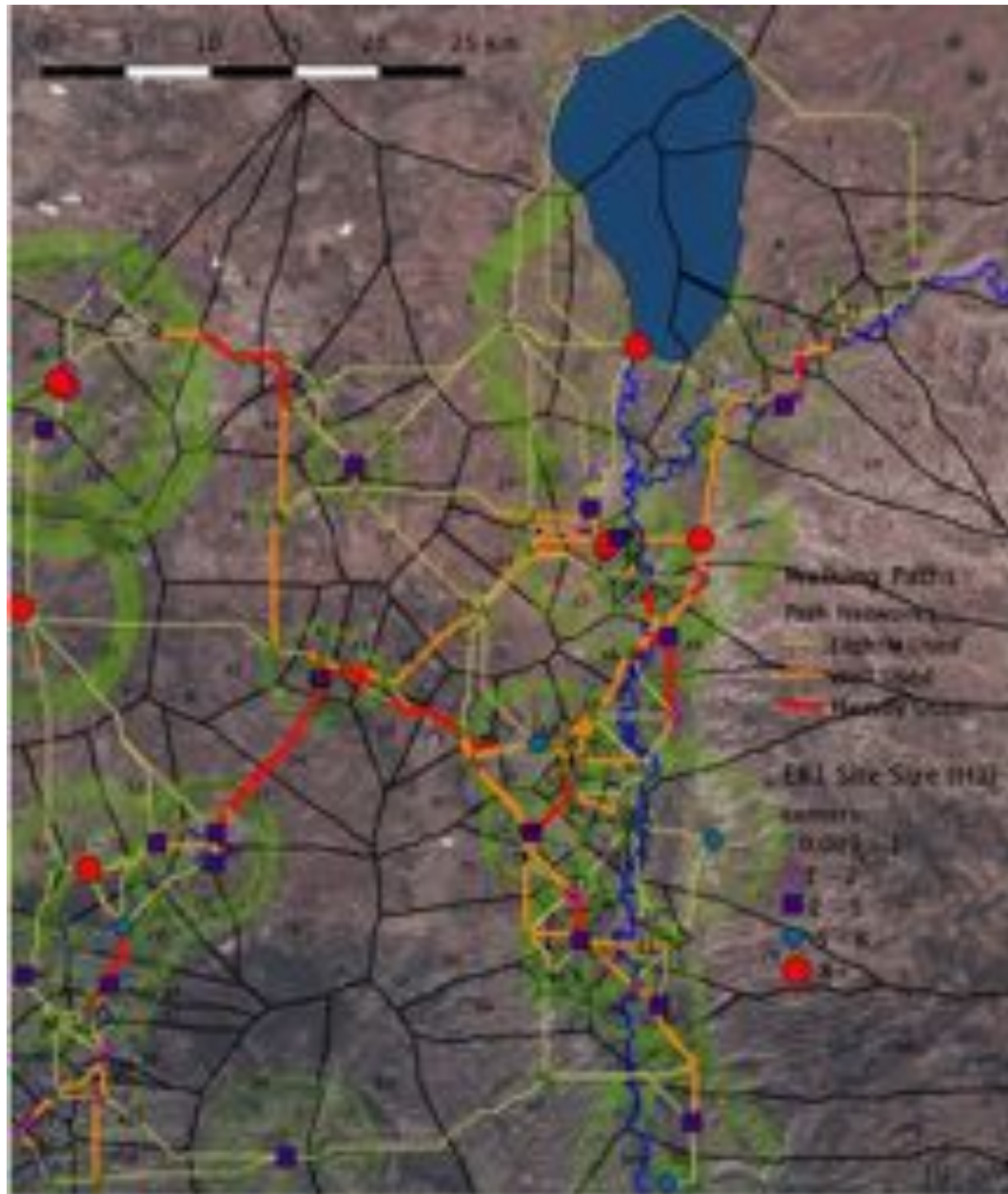


Figure 6.15: Regional map showing weighted path networks, EB I sites (sized), results of XTENT analysis, and catchment analysis results for EEBI maximum population maximum crop extents smallest footprints scenario. Light green indicates land desired for use by one settlement. Dark green indicates land desired for use by 2 settlements. Red and Yellow indicate several sites desiring the same land.

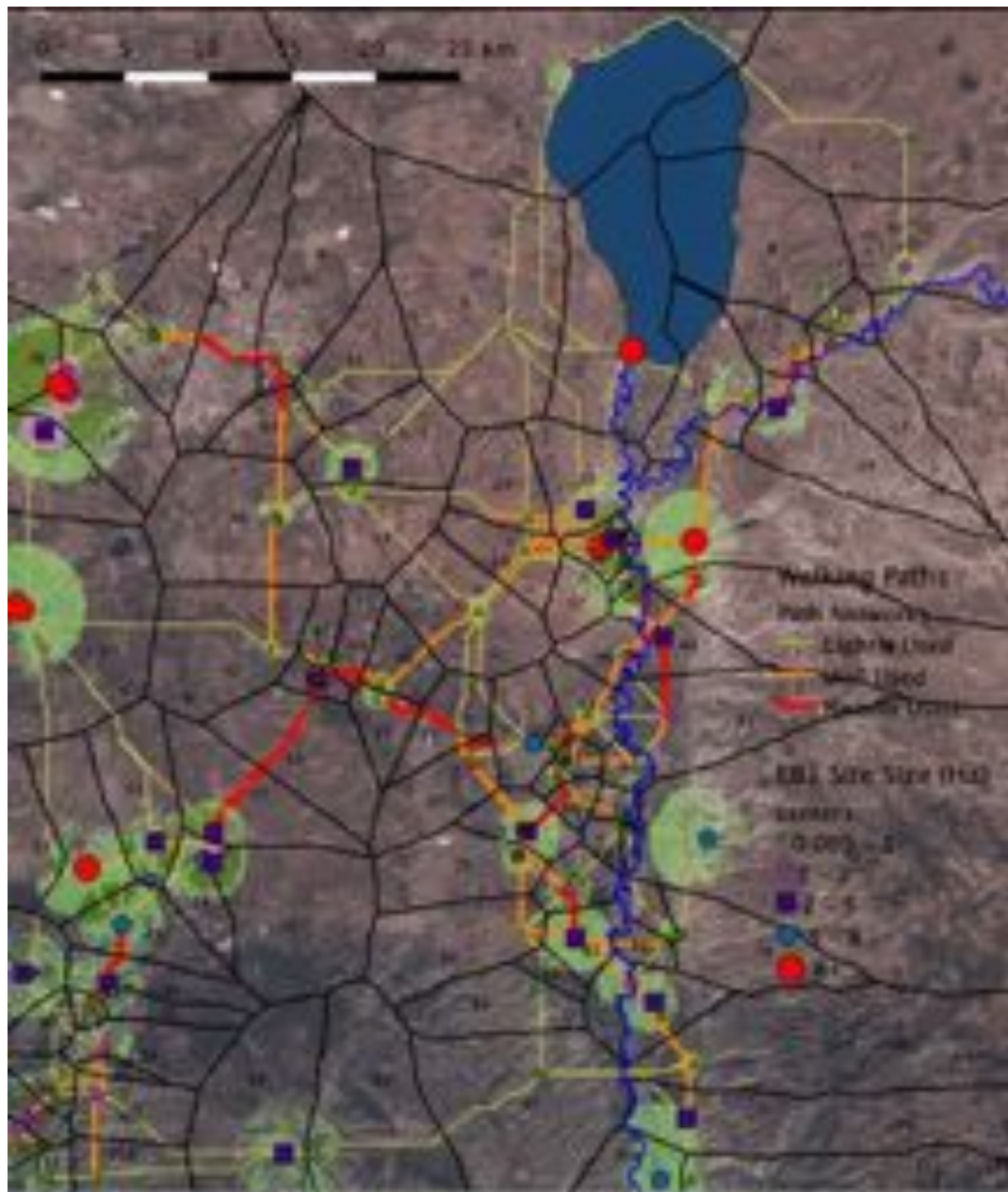


Figure 6.16: Regional map showing weighted path networks, EB I sites (sized), results of XTENT analysis, and catchment analysis results for EEBI minimum population maximum crop extents largest footprints scenario. Light green indicates land desired for use by one settlement. Dark green indicates land desired for use by 2 settlements. Red and Yellow indicate several sites desiring the same land.



Figure 6.17: Regional map showing weighted path networks, EB I sites (sized), results of XTENT analysis, and catchment analysis results for EEBI maximum population maximum crop extents largest footprints scenario. Light green indicates land desired for use by one settlement. Dark green indicates land desired for use by 2 settlements. Red and Yellow indicate several sites desiring the same land.

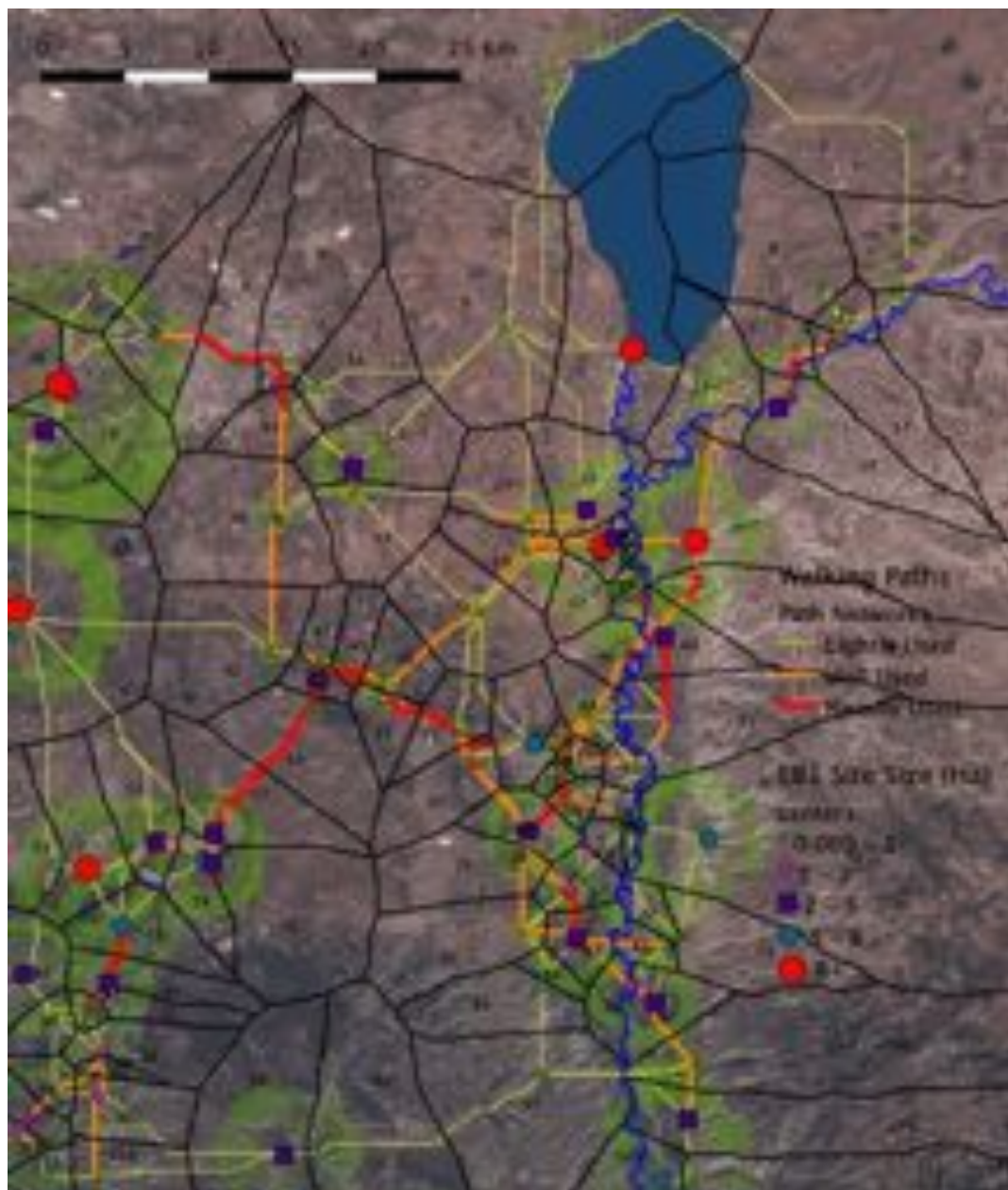


Figure 6.18: Regional map showing weighted path networks, EB I sites (sized), results of XTENT analysis, and catchment analysis results for EEBI minimum population maximum grazing extents largest footprints scenario. Light green indicates land desired for use by one settlement. Dark green indicates land desired for use by 2 settlements. Red and Yellow indicate several sites desiring the same land.

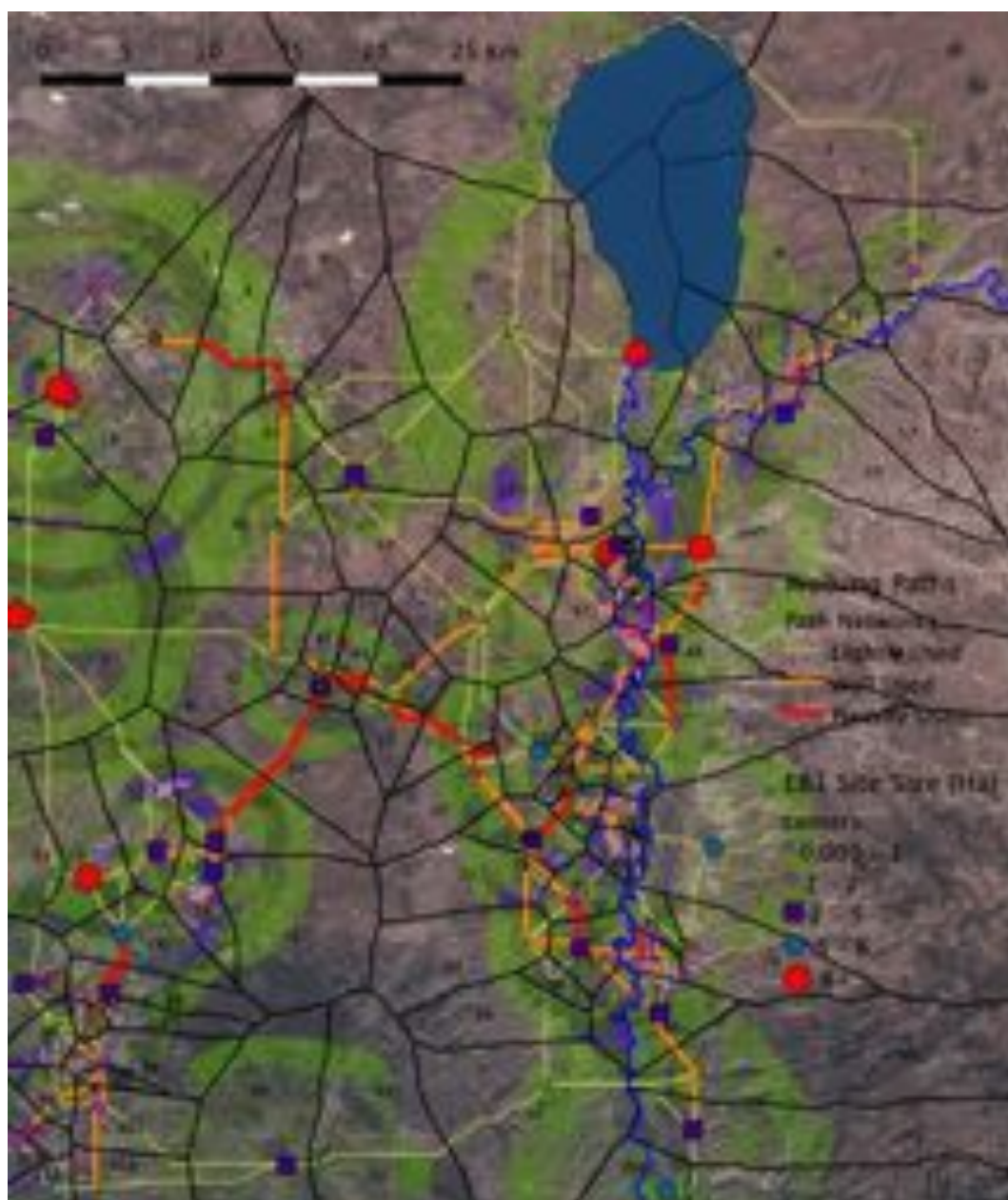


Figure 6.19: Regional map showing weighted path networks, EB I sites (sized), results of XTENT analysis, and catchment analysis results for EEBI maximum population maximum crop extents largest footprints scenario. Light green indicates land desired for use by one settlement. Dark green indicates land desired for use by 2 settlements. Red and Yellow indicate several sites desiring the same land.

6.6 Applying Networks to Shuna

“Site catchment analysis probably gives us a good rough sketch of the resources within walking distance of a village. What it does not give us is a sketch of the network of interrelationships by which those resources were manipulated.”

– Rossmann 1976, 103

The results of the network analysis can be applied to some of the results presented in Chapter 5, where several different landuse scenarios were presented. In particular, the larger population levels for EEBI and LEBI Shuna will benefit because of the potentially large amounts of land required, and the potential for competition from neighbouring sites for land. For example, as shown in Figure 5.100 on page 241, which depicts the catchments calculated for EEBI Maximum Population (3000), Meat 30%, the footprints are very large, and very variable depending on the dairy utilisation and animal foddering factors.

Once the regional path network has been established, and subsistence landuse catchments have been generated for each site in the network, it is possible to do much deeper analysis considering the subsistence needs of (and potential production of) each site in the network. Given the *massive* amounts of data required to properly set up each of the EBA I sites in the southern Levant (from data from the palaeobotanical and faunal records) for LA to perform the modelling in a robust fashion, the huge amount of computational time required, plus the time constraints placed on this research, it has not been possible to take this analysis beyond the point to which it has been presented. Further discussion of the potential of regional route networks in conjunction with subsistence landuse catchment is discussed in the following (final) chapter of this thesis.

Because there was a suspected subsistence strategy change from the EEBI to the LEBI the early EB I diet has been used here to see if the subsistence strategies linked to an EEBI diet could support higher population levels seen in the LEBI.

As evident in 6.16, which illustrates minimum population estimates and subsistence strategy scenario producing maximum cropland requirements there is little overlap and there should be no problems maintaining this population with this scenario. Taking the population level to a higher level however results in a much different picture as evident in 6.17. Here we see the entire Jordan valley dedicated to cultivation and crop production. Even when looking at the maximum population with a subsistence strategy known to produce a minimum area of cropland requirement as in 6.17 the majority of the arable land in the Jordan valley is being intensively cultivated.

Minimum populations and a subsistence strategy which produces the smallest cropland requirements as in Figure 6.16 shows very little overlap suggesting that this combi-

nation of population and subsistence strategies could have been successfully employed. Once grazing land requirements are considered in this regional view it becomes clear how much of the region was required for food production to sustain the growing population. Looking at Figure 6.19 in particular there is very little room for expansion therefore subsistence strategy change would be required in order for population levels to increase much beyond this level. Again looking at the minimum population the subsistence strategies which produce the smallest footprints (e.g. Figures 6.14 to 6.15) suggest there would have been ample room for expansion for agricultural lands to accommodate and increasing population without having to change subsistence strategies.

One other aspect visibly apparent from these regional maps is that land used for grazing in the lower population scenarios would have had to be converted to use as cropland as population levels increased. If this were to happen simultaneously to settlements near each other, there would be potential for increased conflict over landuse; because grazing land is less intensively used in comparison to cultivated cropland there is less visible signs of control / ownership than there would be with cultivated land thus if one settlement began cultivating land that previously another site had used for grazing it would be immediately apparent and arrangements/agreements would have been necessary to avoid conflict.

As populations grew and the amount of cultivated land continued to expand outwards from the settlements fewer choices would have been available to the shepherds for grazing their herds. This could have negative consequences if the variety of ecological zones in their grazing range was reduced as it would have increased vulnerability to fluctuations in weather patterns. Restricted access to grazing lands could have triggered an increased reliance on the use of fodder to lessen the vulnerabilities associated with access to fewer ecological zones. Another effect of large populations using an EEBl subsistence strategy is that grazing lands are becoming increasingly distant in some cases reaching nearly 20 kms which is a substantial distance to move a herd. By the time the overlapping grazing land is relocated to areas which are not being used for either grazing or under-cultivation by other settlements the potential for population increases was essentially limited to this level.

Because grazing land requires no human input for use, and is only ever used when there is grass available for the animals to eat, grazing areas would have a low and ephemeral human presence in comparison to land which is under cultivation. Taking this into account and also considering that grazing lands are located further away from the security of the settlement and its associated busy cropland, it is the grazing lands which would be easiest for other settlements to encroach upon. People could not lightly crop another settlements cultivated land without being discovered. It would be possible however to lightly graze another settlements distant grazing lands without it being discovered. Cultivated land is obvious grazing land is not. The mere fact that the land

is cultivated gives a powerful image of ownership and control of that land.

As populations increased the 'obvious' cropland would have been much less likely to be a source of ongoing conflict in comparison to the more distant less intensively used grazing lands. In particular during times of drought scenarios such as those represented in 6.17 would almost certainly resulted in many problems in this regard. To lessen the problems associated with subsistence strategies that require large amounts of cultivated land and extensive grazing lands subsistence strategies could be adjusted, as was illustrated throughout Chapter 5. Changes, such as an increased use of dairy products, the foddering of animals to reduce grazing land requirements and less meat in the diet, (as is suggested by the LEBI archaeological evidence at Shuna see Figure 5.10 on page 123) would have accommodated higher populations than the early EB I scenarios used in this chapter.

The clustering of large sites would not be problematic if there was no environmental problems like drought because as discussed above the land under cultivation would have had to have been in use under arrangement. It does mean however, that the majority of land in and around the settlements was increasingly dedicated to use as crop land as population levels increased. Once all of the available land was in use as crop land (which seems to be the case in the Jordan valley given the scenarios shown 6.17 for example) it would seem likely that these communities would have benefited from some kind of arrangement with settlements outside of the crop catchment areas required by these large settlements; thus, it would be unsurprising to see many smaller sites in areas less amenable to agriculture that could provide the animal products that the larger centres have become less able to produce themselves. This tendency is apparent in these maps, which lends support to this possibility.

It is easy to imagine grazing areas becoming more distant from the settlement as a result of increasing cultivation, with shepherds taking their families with them into these distant areas and establishing a small settlement. This strategy would have made it easier for them to make their living, while reinforcing their control/ownership of the area with their increased presence. This could have been a development of convenience and could easily have developed into a system which some might recognise as being urban.

As croplands were expanding, and before the majority of arable land was being cultivated prospective cropland in use by shepherds for grazing would have been more important to maintain control/ownership over than areas which were less suitable for agricultural use. One might expect therefore to see indications of people attempting to mark these areas in such a way that they would have been clearly identified as belonging to them. A small settlement would be a good way of doing this as it not only does this by a physical structure but also a human presence. If the cropland continued to expand to accommodate the increasing population and these small settlements became

surrounded by the cropland (pushing grazing land further away) it would again have been beneficial to leave that settlement and start a new one in the now more distant grazing area for the same reasons as before. This might in part explain why the Early Bronze Age saw a cycle of settlements appearing and subsequently being abandoned.

Especially in areas where cropland supplying the larger settlements were in proximity to each other it is reasonable to think that these areas would have made more effort to clearly mark their territories and field boundaries, terraces etc. might have been important. The route network is much more complex and dense in areas of which are under intensive agricultural use, clearly they are more complex simply because of the number of sites but it is clear from the maps that the majority of these sites creating this increased complexity are small, with most being under 1 hectare in size.

It is interesting to note that there appears to be more small sites in the Jordan valley clustered around large sites than elsewhere in the region. This could be due to the geographical constraints of the valley and the river limiting access to the growing need for arable land. It is interesting to see that there appears to be more medium sized sites clustered around the large sites in the west than there are in the Jordan valley where we see a lot more small sites clustered around the medium to large sites. To the east of the Jordan river it appears there would be less potential for conflict and competition for land, but this could be a result of the natural boundary formed by the Jordan River or due to the abrupt change in landscape experienced east of the Jordan valley. Additionally, it could be partly attributed to the fact that, in this preliminary regional study, it has not been possible to include all of the sites east of the Jordan valley due to lack of data. These issues mean that this observation is made with a certain degree of caution. Similar observations are noticeable however, in the region to the south and west of Lake Tiberias.

To conclude, an examination of these scenarios at this regional level would seem to suggest that the subsistence strategies suggested from an early EB I diet which was arguably more dependent on primary animal products would not have been sustainable with a high population. I am suggesting therefore that because population levels did increase a subsistence strategy change would have been necessary.

The following final chapter of this thesis discusses how this research has addressed the research questions posed in Chapter One. Additionally, it presents the future potential development of this work, not only in relation to the analysis presented in Chapters 5 and 6, but also for additional applications and enhancements to the methodology (and Landuse Analyst in particular).